

UST- KHAIRYUZOVO, Russian Federation

WMO: 322870

Lat: 57.095N

Lon: 156.738E

Elev: 8

StdP: 101.23

Time Zone: 12.00 (E12)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.5	-25.8	-31.2	0.2	-28.3	-28.5	0.3	-25.5	12.4	-8.7	11.7	-8.3	2.6	130	0.809

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.4	19.4	15.9	17.7	14.7	16.4	13.9	16.6	18.5	15.3	16.9	14.3	15.7	4.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	11.2	17.3	14.5	10.3	16.1	13.5	9.7	15.2	46.7	18.8	43.0	17.2	40.2	15.9	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.9	8.7	-32.3	22.5	2.2	1.7	-33.9	23.7	-35.2	24.7	-36.5	25.7	-38.1	26.9		
(5)			-32.4	18.7	2.2	1.5	-34.0	19.8	-35.3	20.7	-36.5	21.5	-38.1	22.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.2	-12.9	-13.2	-8.5	-2.5	3.6	8.5	12.5	12.9	8.4	2.5	-4.1	-9.9	(6)
(7)		DBStd	10.29	6.95	6.98	6.17	4.34	2.68	2.56	2.38	2.33	2.66	3.29	4.70	6.18	(7)
(8)		HDD10.0	3908	711	651	574	374	199	58	4	5	61	232	422	616	(8)
(9)		HDD18.3	6747	969	884	833	624	457	296	182	169	297	491	672	874	(9)
(10)		CDD10.0	203	0	0	0	0	1	13	82	94	14	0	0	0	(10)
(11)		CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	3.9	3.8	3.7	3.9	4.2	4.2	4.0	3.7	4.0	3.8	3.8	4.1	3.8	(14)
(15)	Precipitation	PrecAvg	557	23	16	16	24	32	46	64	78	70	80	69	41	(15)
(16)		PrecMax	881	60	56	49	50	81	146	173	171	200	198	138	127	(16)
(17)		PrecMin	336	3	0	1	4	4	0	3	7	11	13	25	6	(17)
(18)		PrecStd	136	16	13	11	13	19	32	45	45	40	37	30	29	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	4.3	5.1	9.5	15.5	20.4	22.6	21.0	17.1	12.3	7.1	6.1	(19)
(20)			MCWB	-0.2	0.7	1.3	4.5	9.7	12.8	16.7	17.3	14.2	7.7	3.8	3.5	(20)
(21)		2%	DB	1.4	1.9	2.9	6.7	12.4	16.4	19.7	19.3	14.9	10.1	4.9	2.6	(21)
(22)			MCWB	-1.4	-0.8	-0.4	3.0	7.8	11.7	16.0	16.4	12.3	6.9	2.2	-0.2	(22)
(23)		5%	DB	-0.3	0.1	1.3	4.7	10.3	14.6	17.9	17.7	13.6	8.8	3.1	0.7	(23)
(24)			MCWB	-2.5	-2.4	-1.2	1.9	6.7	10.9	14.8	15.3	11.6	6.5	1.1	-1.6	(24)
(25)		10%	DB	-2.5	-2.8	-0.4	3.0	8.2	13.0	16.4	16.4	12.6	7.4	1.6	-1.3	(25)
(26)			MCWB	-4.1	-4.1	-2.3	0.9	5.6	10.0	14.0	14.3	10.9	5.5	0.2	-2.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.8	1.5	5.5	10.3	13.8	18.3	18.7	15.5	9.1	4.4	3.5	(27)
(28)			MCDWB	2.7	3.7	4.9	8.0	14.2	17.3	19.6	19.8	16.3	10.6	5.7	6.0	(28)
(29)		2%	WB	-1.3	-0.7	0.0	3.5	8.2	12.3	16.5	17.0	13.3	7.8	2.5	0.2	(29)
(30)			MCDWB	1.0	1.9	2.4	6.1	11.5	15.4	18.9	18.6	14.2	9.4	4.5	2.3	(30)
(31)		5%	WB	-2.5	-2.4	-1.1	2.3	6.9	11.2	15.3	15.7	12.1	6.8	1.3	-1.4	(31)
(32)			MCDWB	-0.3	-0.3	1.0	4.4	9.9	13.7	17.3	17.2	13.1	8.2	2.7	0.5	(32)
(33)		10%	WB	-4.1	-4.2	-2.3	1.1	5.7	10.3	14.3	14.6	11.2	5.8	0.2	-2.9	(33)
(34)			MCDWB	-2.6	-2.6	-0.5	2.7	8.1	12.6	16.1	15.8	12.3	7.4	1.5	-1.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	7.5	7.6	6.3	6.3	6.3	5.8	5.4	6.4	5.2	5.1	6.7	(35)
(36)			MCDBR	6.9	7.1	7.3	8.2	10.4	9.8	8.4	8.0	7.3	7.4	5.9	6.4	(36)
(37)			MCWBR	5.4	5.4	5.3	5.4	6.5	5.8	5.2	4.9	5.3	5.2	4.4	5.0	(37)
(38)		5% WB	MCDBR	6.8	7.1	6.9	7.2	9.7	8.3	7.5	6.7	6.0	6.4	5.5	6.3	(38)
(39)			MCWBR	5.4	5.7	5.2	4.9	6.4	5.6	5.1	4.8	5.0	5.2	4.3	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.261	0.298	0.344	0.425	0.467	0.436	0.455	0.427	0.361	0.349	0.301	0.264	(40)	
(41)		taud	2.134	2.107	2.030	1.898	1.958	2.170	2.150	2.227	2.395	2.241	2.117	2.052	(41)	
(42)		Ebn at Noon	606	732	788	771	767	805	777	773	784	665	527	468	(42)	
(43)		Edh at Noon	54	87	123	165	167	137	137	119	87	77	56	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.57	1.42	2.90	4.14	4.95	5.14	4.81	3.48	2.58	1.38	0.55	0.33	(44)	
(45)		RadStd	0.06	0.13	0.28	0.61	0.52	0.64	0.50	0.39	0.24	0.14	0.07	0.03	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page